GROCERY SHOP TRANSACTION & BUSINESS PERFORMANCE ANALYSIS

Data Quality Investigation, Sales Analysis, Business Insight and Recommendations

An end-to-end data analytics project using Microsoft Excel, MySQL and Power BI

Executive Summary

This project examines real-world transaction data from a local grocery shop to assess recorded customer sales performance, transaction patterns, and financial activity. The objective was to transform raw M-PESA transaction records into meaningful business insights that could support performance monitoring and informed decision-making.

The analysis followed an end-to-end workflow using Microsoft Excel, MySQL, and Power BI. Excel Power Query was used to combine, clean, and prepare transaction records. MySQL was used to query the dataset, calculate performance metrics, investigate data-quality issues, and validate transaction totals. Power BI was used to present key performance indicators and interactive visualizations through a business overview dashboard and a detailed analysis page.

The dataset contained 18,101 transaction records covering 29 August 2023 to 29 August 2026. The records included KES 687,683 in total incoming payments and KES 685,610 in total outgoing payments. Customer sales accounted for KES 681,132 in recorded incoming payments.

The analysis identified a substantial increase in recorded customer sales from KES 81,577 in 2024 to KES 336,292 in 2025, representing an increase of approximately 312.5%. Customer-sale transaction volume also increased, reaching 8,394 transactions in 2025. Monthly sales analysis highlighted August 2025 as the strongest-performing month, with KES 36,498 in recorded customer sales, followed by July at KES 36,310 and October at KES 33,613. These findings provide a basis for further investigation into sales patterns and business operations.

Data-quality checks examined missing values in key fields, negative amounts, repeated receipt numbers, zero-value transactions, transaction classifications, and consistency between the SQL analysis table and the Power BI reporting view. Exceptions were documented, and records were retained where the available evidence did not justify changing their values or classifications.

The project demonstrates how data preparation, SQL analysis, data visualization, and business interpretation can be combined to support practical decision-making. However, M-PESA transaction records alone do not establish the shop's overall profitability or explain the causes of changes in sales. Additional information, including product-level sales, inventory, purchase costs, and operating expenses, would support a more comprehensive assessment of business performance.

1. Project Objectives and Scope

1.1 Project Objectives

The main objective of this project was to analyze transaction data from a local grocery shop to understand its recorded sales performance, transaction patterns, and financial activity.

The specific objectives were to:

- Prepare and clean raw M-PESA transaction records using Microsoft Excel and Power Query.
- Use MySQL to query transaction data, calculate key performance metrics, and analyze sales trends.
- Examine transaction categories to distinguish customer sales from financing, supplier payments, owner withdrawals, and transaction charges.
- Investigate data-quality issues and validate the consistency of transaction records and reporting totals.
- Develop an interactive Power BI dashboard to communicate key findings and support business performance monitoring.
- Translate analytical findings into practical recommendations for the business.

1.2 Project Scope

The analysis covered 18,101 M-PESA transaction records dated from 29 August 2023 to 29 August 2026. The work focused on recorded customer sales, transaction volumes, average transaction values, monthly and annual sales patterns, and the movement of money into and out of the account.

The project used Microsoft Excel for data preparation, MySQL for querying and validation, and Microsoft Power BI for visualization and reporting.

The scope was limited to the information available in the supplied M-PESA statement. Product-level sales, inventory records, purchase costs, operating expenses, and transactions conducted through other accounts were not available for this analysis. Therefore, the project assesses recorded transaction activity and sales patterns rather than calculating complete business profitability.

1.3 Key Analytical Questions

The project was guided by the following questions:

1. How did recorded customer sales change over the reporting period?
2. Which months recorded relatively high or low customer sales?
3. How did customer transaction volume and average transaction value change over time?
4. How were incoming and outgoing payments distributed across transaction categories?
5. What data-quality issues required investigation, and how reliable were the prepared reporting totals?
6. How can the findings support better business monitoring and decision-making?

2. Methodology and Tools

2.1 Overview of the Analytical Workflow

The project followed a structured data analytics workflow, beginning with raw M-PESA transaction records and progressing through data preparation, SQL analysis, visualization, and business interpretation. Each tool served a specific purpose in preparing the data, validating results, identifying patterns, and communicating findings.

2.2 Data Preparation Using Microsoft Excel

Microsoft Excel and Power Query were used to combine transaction data from four source files into a consolidated dataset. The preparation process included reviewing the source fields, standardizing data types, converting transaction dates, cleaning blank rows, and assigning transaction categories.

The resulting dataset was organized for further analysis and included transaction details, dates, incoming and outgoing amounts, balances, statement periods, and transaction categories. Data-quality checks were also documented to record unusual transactions and issues requiring further investigation.

2.3 Data Analysis Using MySQL

MySQL was used to store, query, summarize, and validate the prepared transaction data. SQL queries were developed to calculate transaction counts and monetary totals, analyze customer sales by month and year, examine average transaction values, and compare performance across reporting periods.

Grouping, conditional logic, common table expressions, date functions, and window functions were used where appropriate to investigate patterns and calculate analytical metrics. Additional queries examined transaction categories, repeated receipt numbers, unusual descriptions, and other data-quality concerns.

The SQL analysis also supported reconciliation between the main transaction table and the reporting view prepared for Power BI.

2.4 Data Visualization Using Microsoft Power BI

Microsoft Power BI was used to build an interactive dashboard for presenting the main findings. The report was organized into a Business Overview page and a Detailed Analysis page.

The dashboard included key performance indicators and visualizations covering customer sales, transaction volume, average transaction value, monthly sales trends, annual performance, and transaction categories. A year slicer enabled the user to explore performance across different reporting periods.

The purpose of the dashboard was to make the analytical results easier to interpret and support ongoing business performance monitoring.

2.5 Business Interpretation and Recommendations

The final stage involved interpreting the results in the context of the grocery shop's recorded transaction activity. The analysis identified sales patterns, changes in transaction volume, differences between financial transaction categories, and data-quality limitations.

Recommendations were developed from the available evidence, with care taken to distinguish observed results from possible explanations. Because the dataset was limited to the available M-PESA statement, conclusions about profitability, product demand, and the causes of sales changes require additional business information.

3. Data Quality Investigation and Assessment

This section documents the data-quality checks performed on the prepared transaction dataset. The investigation examined missing values, negative amounts, repeated receipt numbers, zero-value transactions, transaction classifications, and consistency between the SQL analysis table and the Power BI reporting view. Findings and exceptions were documented to support transparency and explain how the data was handled before reporting.

Project: Grocery Shop Transaction & Business Performance Analysis
Tool: MySQL
Table: grocery_transactions_full

3.1 Initial Data Quality Checks

Data-quality check	Result	Interpretation	Action
Missing receipt numbers	0	No NULL receipt numbers found	No action required
Missing transaction dates	0	No NULL transaction dates found	No action required
Missing paid-in amounts	0	No NULL paid-in values found	No action required
Missing withdrawn amounts	0	No NULL withdrawn values found	No action required

Missing transaction categories	0	No NULL categories found	No action required
Data-quality check	**Result**	**Interpretation**	**Action**
Negative paid-in amounts	0	No negative values found	No action required
Negative withdrawn amounts	0	No negative values found	No action required
Exact duplicate groups across eight fields	0	No exact matches found using those fields	No action required
Records with both amounts equal to zero	13	Requires investigation	Retain and flag verification

3.2 Receipt Number and Zero-Value Transactions

Finding: Repeated receipt numbers

Repeated receipt numbers were identified. However, records sharing a receipt number had different descriptions, amounts, or categories. Repeated receipt numbers alone do not establish that transactions are duplicates.

Decision: No records were deleted based on receipt-number repetition.

Finding: Zero-amount merchant payment charges

Thirteen records categorized as Charges / Fees contained zero values for both Paid In (KES) and Withdrawn (KES).

A sample record, TE42FKPFJ8, was checked against the original Excel data. The zero values were already present in Excel, confirming that the SQL import did not introduce them.

Current status: The cause of the zero amounts remains unconfirmed.

Action: Retain the records and flag them for verification against the original M-PESA statement. Do not invent replacement amounts.

Potential business impact: If these records represent charges whose amounts were omitted from the source data, reported expenses may be understated.

3.3 Supplier / Business Payment Investigation

Checks performed: Transaction counts, unique descriptions, missing descriptions, withdrawal totals, and grouped description totals.

Results: The category contains 116 transactions across 13 unique descriptions, totalling KES 171,834 in withdrawals. All grouped counts and amounts reconcile with the category summary. No descriptions are missing or blank.

Business validation: The shop owner confirmed that the reviewed payments are for business supplies.

Conclusion: No data-quality issue requiring category reclassification was identified in this investigation. The WITH ROLLUP output's [NULL description] label represents the grand total, not a missing-description record.

3.4 Financing Investigation

Confirmed: The Financing category contains 145 transactions, KES 6,551 paid in, and KES 29,239 withdrawn.

Observed: The descriptions include overdraft repayments, loan disbursements, and overdraft-related transfers.

Pending clarification: Whether transfers involving Biashara Overdraft use only borrowed funds or a combination of borrowed funds and the shop's available balance.

Action: Keep the original classifications until the funding arrangement is confirmed.

3.5 Owner Withdrawal Investigation

Checks performed: Grouped transactions by description, checked transaction counts and withdrawal totals, and investigated the record with an unusual description.

Results:

- Total transactions: 318
- Total withdrawals: KES 481,282
- “Settle funds to - BILLY ODHIAMBO”: 317 transactions totaling KES 480,882.
- “Withdraw Funds from NEWA PROPERTIES Ltd”: 1 transaction totaling KES 400, dated 5 November 2023.

Finding: The KES 400 transaction has a different description from the other records in this category. However, the transaction is too old to verify its business purpose.

Decision: Retain the transaction under Owner Withdrawal based on its existing description and classification. Do not reclassify it without supporting evidence.

Limitation: The ultimate purpose of the KES 400 withdrawal could not be independently confirmed.

3.6 Charges / Fees Investigation

Checks performed: Grouped records by description, reconciled transaction counts and withdrawal totals, inspected zero-value records, and matched their receipt numbers to transactions in other categories.

Results:

- Total transactions: 264
- Unique descriptions: 5
- Total recorded withdrawals: KES 3,255
- Missing or blank descriptions: 0
- Zero-value fee records: 13
- All 13 zero-value fee records share receipt numbers with corresponding Customer Sale records.
- Matching customer-sale amounts range from KES 10 to KES 140.

Finding: The 13 fee records have zero values in both the Paid In and Withdrawn columns. Their matching receipt numbers provide evidence of an association with customer-sale transactions, but the actual fee amounts cannot be established from the available fields.

Decision: Retain the original records and zero values. Do not estimate or impute fee amounts. Verify against the original M-PESA statement or a more detailed transaction report if available.

Limitation: The actual amounts, if any, associated with these fee records remain unverified.

3.7 Customer Sale Description Review

The final review identified three transactions totaling KES 30 with the description pattern “Business Buy Goods by 901670 - Bonga Everywhere Services via API.” Each transaction recorded KES 10 paid in and KES 0 withdrawn.

These descriptions differ from the dominant merchant-payment patterns. The records were retained under the existing Customer Sale category because the available data does not establish a different classification.

Status: Reviewed — unusual description documented; transaction purpose unverified.

The Customer Sale category reconciles to 17,258 transactions and KES 681,132 in incoming payments, with no withdrawals. No records were changed during this review.

3.8 Consolidated Data Quality Assessment

3.8.1 Dataset Overview

The prepared dataset contains 18,101 M-PESA transaction records covering the period from 29 August 2023 to 29 August 2026. Total recorded incoming payments amounted to KES 687,683, while total recorded outgoing payments amounted to KES 685,610. Customer-sale transactions accounted for KES 681,132 in recorded incoming payments.

These figures were reconciled between the main MySQL transaction table and the reporting view used by Power BI.

3.8.2. Data Quality Checks and Results

Check	Result	Status
Missing values in key fields	No missing values found in Receipt No., Transaction Date, Paid In, Withdrawn, or Transaction Category	Passed

Negative monetary values	No negative values found in Paid In or Withdrawn	Passed
Transactions with both monetary fields equal to zero	13 records, all in Charges / Fees	Exception documented
Duplicate receipt numbers	Some receipt numbers appear on multiple transaction rows; however, no exact duplicate combinations were found across all eight fields.	Reviewed
Category reconciliation	Counts and monetary totals reconcile with the overall dataset	Passed
Missing transaction descriptions	No missing or blank descriptions found in the reviewed categories	Passed
Unusual transaction descriptions	Investigated across transaction categories	Reviewed; exceptions documented
SQL table and Power BI view reconciliation	Both contain 18,101 records, KES 687,683 in Paid In and KES 685,610 in Withdrawn	Passed

3.8.3. Overall Financial Reconciliation

Measure	**Total (KES)**
Total Paid In	687,683
Total Withdrawn	685,610
Difference	2,073

The difference between total money received and total money withdrawn is KES 2,073. This represents the net difference between the two transaction totals; it should not automatically be interpreted as profit or cash available.

3.8.4. Category-Level Validation

Transaction Category	Records	Paid In (KES)	Withdrawn (KES)
Customer Sale	17,258	681,132	0
Supplier / Business Payment	116	0	171,834
Financing	145	6,551	29,239
Charges / Fees	264	0	3,255
Owner Withdrawal	318	0	481,282
Total	18,101	687,683	685,610

3.8.5. Exceptions and Limitations

Charges / Fees

Thirteen records contain zero in both Paid In and Withdrawn. These values were present in the original Excel data. The records were retained without estimating missing amounts because the actual fees could not be established from the available statement data.

Owner Withdrawal

One KES 400 transaction dated 5 November 2023 has a description that differs from the dominant owner-settlement pattern. It was retained in the existing category because the transaction was too old to verify**.**

Financing

The records include Biashara overdraft repayments, loan disbursements, overdraft withdrawals, and overdraft-related transfers. The precise funding composition of transactions described as “Funds Transfer with Biashara Overdraft” remains unconfirmed. The existing classifications were retained pending clarification.

Customer Sale

Three transactions totaling KES 30 use the description “Business Buy Goods by 901670 - Bonga Everywhere Services via API.” They were retained under Customer Sale because the available data did not establish a different purpose. Some owner-associated incoming payments may represent purchases from the shop, but individual transaction purposes were not independently verified.

Repeated Receipt Numbers

Repeated receipt numbers were not automatically treated as duplicate transactions because a receipt number can appear on separate transaction lines, such as a payment and a related fee. No exact duplicate combinations were identified by the grouping check performed.

3.8.6. Conclusion

The dataset has undergone checks for missing values, negative monetary amounts, zero-value transactions, repeated receipt numbers, category consistency, and unusual transaction descriptions.

The overall transaction count and monetary totals reconcile across the five categories. Identified exceptions have been retained and documented rather than changed based on assumptions.

The dataset is suitable to proceed to the next stage of analysis, with the documented limitations kept in mind. Before final publication, any outstanding Financing clarification should be recorded if it becomes available.

Business Insights and Recommendations:

1. Customer Sales Growth and Business Performance

The analysis revealed a substantial increase in recorded customer sales between 2024 and 2025. Customer sales rose from 81,577 in 2024 to KES 336,292 in 2025, representing an increase of approximately 312.5%.

This growth indicates a significant increase in recorded customer-sale inflows during 2025. Identifying the factors associated with this increase could help the business owner understand which activities, customer purchasing patterns, or operational changes coincided with stronger performance.

Business implication: The strong performance recorded in 2025 provides an opportunity for the business owner to investigate successful sales periods and identify practices that may be worth maintaining or improving.

Recommendation: The business owner should review monthly sales trends, identify the strongest-performing months, and investigate possible reasons for increased sales. Consistent monthly monitoring would also help track performance and support more informed business decisions.

Data limitation: The dataset covers 29 August 2023, to 29 August 2026. Therefore, the 2023 and 2026 figures represent partial years and should not be compared directly with complete calendar-year totals. Although the analysis identifies a substantial increase in recorded customer sales, the transaction data alone does not establish the cause of that growth.

2. Monthly Sales Performance and Variations

The monthly sales analysis revealed variations in customer sales performance across the reporting period. August 2025 recorded the highest monthly customer sales, at KES 36,498, followed by July 2025, at KES 36,310. October 2025 also recorded strong sales of KES 33,613.

These findings show that several months in 2025 recorded particularly strong customer sales. However, the available transaction data does not establish why these months performed well. Further investigation would be needed to determine whether factors such as customer demand, product availability, pricing, promotions, or other business activities contributed to the results.

Business implication: Identifying high-performing months can help the business owner better understand sales patterns and plan inventory, cash flow, and other operational activities.

Recommendation: The business owner should compare monthly sales across reporting periods, investigate unusually high or low months, and consider seasonal factors alongside stock availability and purchasing records. This could help identify recurring sales patterns and improve inventory and cash-flow planning.

Data limitation: The dataset begins on 29 August 2023, meaning August 2023 contains only a few days of activity. Comparisons involving partial months should therefore be interpreted cautiously.

3. Transaction Categories and Business Cash Flow

The transaction analysis identified several distinct categories of financial activity within the grocery shop's M-PESA statement. These included customer sales, supplier and business payments, financing, owner withdrawals, and transaction charges.

Customer sales accounted for KES 681,132 in recorded incoming payments. Financing transactions accounted for KES 6,551 in incoming funds, while supplier and business payments totalled KES 171,834 in outgoing funds. Owner withdrawals amounted to KES 481,282, and transaction charges totalled KES 3,255.

These findings demonstrate why transaction classification is important when assessing business performance. Incoming financing should not be treated as sales revenue, while owner withdrawals and supplier payments represent different uses of funds rather than customer sales.

Business implication: Separating customer sales from other financial transactions provides a clearer picture of recorded business activity and reduces the risk of overstating sales performance.

Recommendation: The business owner should maintain consistent transaction categories and review customer sales, supplier payments, financing, owner withdrawals, and transaction charges separately. Where possible, business and personal transactions should also be clearly distinguished to improve financial monitoring.

Data limitation: M-PESA transaction records provide evidence of money received and paid through the account, but they do not establish the shop's complete profitability. The analysis does not include all the information required to calculate profit, such as the cost of goods sold, inventory changes, operating expenses, or transactions conducted through other accounts and payment channels.

4. Customer Transaction Trends and Average Transaction Value

The analysis showed that customer transaction volume increased substantially between 2023 and 2025. Recorded customer transactions rose from 702 in 2023 to 2,042 in 2024 and 8,394 in 2025. The 2026 dataset recorded 6,120 customer transactions through 29 August 2023.

Average customer transaction value also changed over the reporting period. The annual averages were approximately KES 33.65 in 2023, KES 39.95 in 2024, KES 40.06 in 2025, and KES 39.16 in 2026.

These results indicate that the strong customer sales performance recorded in 2025 coincided with a substantial increase in transaction volume, while the average transaction value remained relatively stable between 2024 and 2026.

Business implication: The increase in transaction volume suggests that the shop processed more customer purchases in 2025. Since the average transaction value remained around KES 39–40 during 2024–2026, the increase in sales was associated primarily with more recorded transactions rather than a large increase in the average amount per transaction.

Recommendation: The business owner should monitor customer transaction counts and average transaction value together. This can help distinguish whether sales changes are associated with transaction volume, average purchase amounts, or both. Any strategy intended to increase average purchase value should be evaluated using subsequent transaction data.

Data limitation: The dataset begins on August 29, 2023, and ends on August 29, 2026. Therefore, the 2023 and 2026 figures cover partial years and should not be directly compared with complete-year totals without accounting for the different reporting periods. Transaction counts also represent recorded customer-sale transactions, not necessarily unique individual customers.

5. Data Quality and Reliability

Data-quality checks were conducted during the preparation and analysis of the grocery shop's M-PESA transaction records. These checks covered missing values in key fields, negative monetary amounts, repeated receipt numbers, zero-value transactions, transaction descriptions, category consistency, and reconciliation between the SQL analysis table and the Power BI reporting view.

The checks identified repeated receipt numbers, but no exact duplicate combinations were found across the eight fields tested. Thirteen records contained zero in both Paid In and Withdrawn, and these were retained because their actual values could not be established from the available data. Unusual transaction descriptions and other classification questions were also documented for further review where necessary.

The SQL analysis table and Power BI reporting view were also reconciled and returned matching totals of 18,101 records, KES 687,683 in incoming payments, and KES 685,610 in outgoing payments. This confirmed consistency between the two reporting datasets.

Business implication: Data-quality checks and reconciliation improve confidence that the dashboard reflects the underlying transaction records consistently. However, repeated receipt numbers and incomplete descriptions and ambiguous transaction details require careful interpretation.

Recommendation: The business owner should retain transaction records, use consistent payment descriptions, investigate repeated receipt numbers where necessary, and document the reasons for unusual or incomplete transactions. Future analyses should continue to include data-quality checks and reconciliation between source data and reporting outputs.

Data limitation: Matching totals between the SQL table and Power BI view confirm consistency between those two representations, but do not independently prove that every original transaction is accurate or complete Ambiguous transaction details and unverified transaction purposes may still limit the interpretation of some records.

6. Business Recommendations

Based on the findings from the Excel, MySQL, and Power BI analyses, the following recommendations are proposed to support more consistent business monitoring and informed decision-making.

6.1 Monitor Sales Performance Monthly

The business owner should review monthly customer sales and transaction volumes to identify changes in performance. The strong results recorded in 2025, particularly in July, August, and October, provide useful starting points for investigating factors associated with higher sales.

6.2 Improve Inventory Planning

Monthly sales trends should be considered alongside product-level sales and stock records when planning inventory. The owner should identify which products contribute most to sales, monitor stock availability, and use purchasing and inventory records to support restocking decisions.

6.3 Track Transaction Volume and Average Transaction Value

Customer-sale transaction counts and average transaction value should be monitored together. This would help the owner assess whether changes in recorded sales are associated with transaction volume, average purchase amounts, or both.

6.4 Maintain Clear Financial Records

Customer sales should be recorded separately from financing, supplier and business payments, owner withdrawals, and transaction charges. Where possible, business and personal transactions should also be distinguished. This would improve financial monitoring and help prevent cash movements from being mistaken for sales revenue or profit.

6.5 Strengthen Transaction Record-Keeping

The business owner should maintain clear payment descriptions, consistent transaction categories, and supporting records for unusual or incomplete entries. Repeated receipt numbers should be reviewed where necessary rather than automatically treated as duplicate transactions.

6.6 Expand Future Business Analysis

Future analysis should incorporate product-level sales, inventory records, purchase costs, operating expenses, and other relevant financial records where available. These additional data sources would support more detailed assessments of product performance, stock management, costs, and profitability.

6.7 Continue Using Data to Support Decisions

The Excel analysis, MySQL queries, and Power BI dashboard should be updated periodically using consistent reporting periods. Regular reviews would help the owner monitor performance over time and evaluate whether operational changes are associated with the intended outcomes.

Important limitation: These recommendations are based on the transaction data available from the M-PESA statement. They should be considered alongside additional business records and the owner's knowledge of day-to-day operations before major financial or inventory decisions are made.

7. Conclusion and Lessons Learned

This project analyzed real-world transaction data from a local grocery shop to examine recorded customer sales performance, transaction patterns, and financial activity. The analysis followed an end-to-end workflow using Microsoft Excel, MySQL, and Microsoft Power BI.

Excel was used to prepare, combine, and categorize the transaction records and investigate data-quality issues. MySQL supported data validation, transaction classification, and the calculation of business performance metrics. Power BI was used to present key performance indicators and interactive visualizations through a business overview dashboard and a detailed analysis page.

The findings revealed a substantial increase in recorded customer sales from KES 81,577 in 2024 to KES 336,292 in 2025. The analysis also identified differences in monthly sales performance, an increase in recorded customer-sale transaction volume, and a relatively stable average transaction value between 2024 and 2026. These findings provide a basis for further investigation into sales patterns and business operations.

The project demonstrated the importance of validating data, documenting unusual transactions, distinguishing customer sales from other financial activities, and interpreting results within the limits of the available evidence. It also highlighted that M-PESA transaction records alone cannot establish overall profitability or explain the causes of changes in sales.

Through this project, practical skills were developed in data preparation, SQL querying, data-quality assessment, business performance analysis, dashboard development, and communicating evidence-based recommendations. The project also established a foundation for future analysis incorporating product-level sales, inventory, purchase costs, and operating expenses.

Overall, the analysis provides a structured starting point for monitoring the grocery shop's recorded transaction performance. Combining the dashboard with additional operational and financial records would enable a more comprehensive assessment of business performance and support better-informed decisions.